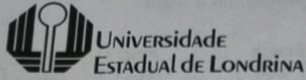


✧ Exercícios teóricos - Compiladores ii ✧

Gabriel Peres de Souza



1COP029/2COP057 - Lista de Exercícios 16

1. ✎ Exercício Teórico: Considere o código C mostrado a seguir:

```
int fib (int n)
{
    if (n == 0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return fib(n-1) + fib(n-2);
}
```

Considere agora a versão em *assembly* de MIPS do código apresentado:

```
fib:    addi    $sp,$sp,-12
        sw      $a0,8($sp)
        sw      $ra,4($sp)
        sw      $s0,0($sp)
        bne     $a0,$zero,$L2  → caso base
        add     $v0,$zero,$zero
        j       $L4
$L2:    addi     $v0,$zero,1 } resolução do
        beq      $v0,$a0,$L4 } caso base
$L3:    addi     $a0,$a0,-1 }
        jal      fib      } return fib(n-1) + fib(n-2)
        add     $s0,$v0,$zero
        addi     $a0,$a0,-1
        jal      fib
        add     $v0,$s0,$v0
$L4:    lw       $s0,0($sp) }
        lw       $ra,4($sp) } encerra
        lw       $a0,8($sp) } programa
        addi     $sp,$sp,12
        jr       $ra
```

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Considere agora a configuração de memória, pilha e registradores apresentada a seguir:

		Pilha/Stack	Registradores
00:	fib:	addi \$sp,\$sp,-12	\$pc: 120
04:		sw \$a0,8(\$sp)	\$ra: 132
08:		sw \$ra,4(\$sp)	\$sp: 1200
12:		sw \$s0,0(\$sp)	\$a0: 5
16:		bne \$a0,\$zero,\$L2	\$v0: 8
20:		add \$v0,\$zero,\$zero	\$s0: 669
24:		j \$L4	\$s1: 666
28:	\$L2:	addi \$v0,\$zero,1	
32:		beq \$v0,\$a0,\$L4	
36:	\$L3:	addi \$a0,\$a0,-1	
40:		jal fib	
44:		add \$s0,\$v0,\$zero	
48:		addi \$a0,\$a0,-1	
52:		jal fib	
56:		add \$v0,\$s0,\$v0	
60:	\$L4:	lw \$s0,0(\$sp)	
64:		lw \$ra,4(\$sp)	
68:		lw \$a0,8(\$sp)	
72:		addi \$sp,\$sp,12	
76:		jr \$ra	
...			
120:		addi \$s0,\$zero,661	
124:		addi \$a0,\$zero,5	
128:		jal fib	
132:		add \$s0,\$v0,\$s0	
136:		lw \$s1,0(\$sp)	
140:		bne \$s0,\$s1,\$ERRO	
144:		lw \$v0,0(\$sp)	
148:		...	
...			
600:	\$ERRO:	j \$ERRO	
...			

Considerando que o registrador \$pc (contador de programa) contém o valor 120, indicando que a instrução armazenada nesse endereço de memória será a próxima a ser executada, mostre a configuração final da pilha e dos registradores quando o registrador \$pc atingir o valor 148.

pilha:	registradores:
1200: 666	\$pc: 148
1196: vazia	\$ra: 132
:	\$sp: 1200
:	\$a0: 5
:	\$v0: 8
:	\$s0: 669
:	\$s1: 666

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LISTA 17 - EXERCICIO 1 - LETRA A

ADDI r2, \$sp, 32

LW r3, 0(r2)

ADDI r4, i, -4

ADD r5, r3, r4

ADDI r6, \$sp, 28

LW r7, 0(r6)

SW r7, 0(r5)

LISTA 17 - EXERCICIO 1 - LETRA B

ADDI r1, \$zero, 32

ADD r2, \$sp, r1

ADDI r3, \$zero, 4

LW r4, 0(r2)

SUB r5, i, r3

ADDI r6, \$zero, 28

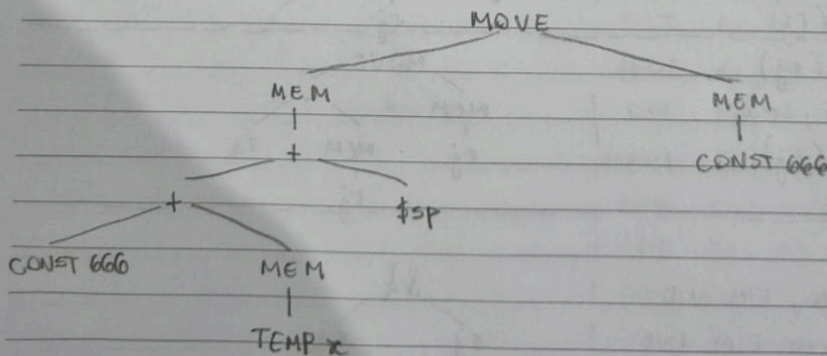
ADD r7, r4, r5

ADD r8, \$sp, r6

LW r9, 0(r8)

SW r9, 0(r7)

LISTA 17 - EXERCICIO 2 - LETRA A



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LISTA 17. EXERCICIO 2. LETRA B

LW r1, 0(x)
 ADDI r2, r1, 666
 ADD r3, r2, \$SP
 LW r4, 0(666)
 SW r4, 0(r2)

LISTA 17. EXERCICIO 2. LETRA C

LW r1, 0(x)
 ADDI r2, \$zero, 666
 ADD r3, r2, r1
 ADD r4, r3, \$SP
 ADDI r5, \$zero, 666
 LW r6, 0(r3)
 SW r6, 0(r4)

LISTA 17. EXERCICIO 3

instruction

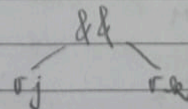
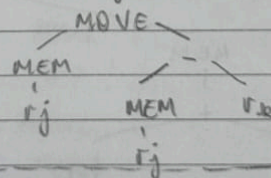
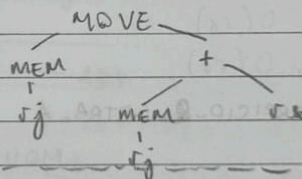
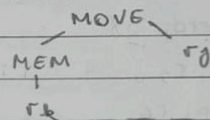
= SW r0, 0(rj)

+= LW ri, 0(rj)
 ADD ri, ri, rk
 SW ri, 0(rj)

-= LW ri, 0(rj)
 SUB ri, ri, rk
 SW ri, 0(rj)

&& LI ri, 0
 beq rj, \$zero, FIM_AND
 beq rk, \$zero, FIM_AND
 LI ri, 1
 FIM_AND:

trees



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Instruction

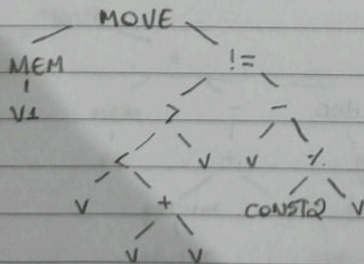
trees

|| LI r_i, 1bne r_j, #zero, FIM_ORbne r_a, #zero, FIM_ORLI r_i, 0

FIM_OR:

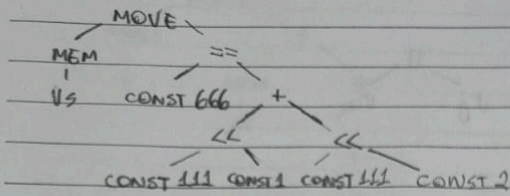
== seq r_i, r_j, r_k!= sne r_i, r_j, r_k< slt r_i, r_j, r_kslti r_i, r_j, c> slt r_i, r_k, r_j>= slt r_i, r_j, r_kxori r_i, r_i, 1<= slt r_i, r_k, r_jxori r_i, r_i, 1

LISTA 17 - EXERCICIO 4 - LETRA A

ADD r₀, v, vSLT r₁, v, r₀SLT r₂, v, r₁ADDI r₃, #zero, 2DIV r₃, vMFHI r₄SUB r₅, v, r₄SNE r₆, r₂, r₅SW r₆, 0(v₄)

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LISTA 17, EXERCICIO 4 - LETRA E



ADDI r0, \$zero, 111

ADDI r1, \$zero, 1

ADDI r2, \$zero, 111

ADDI r3, \$zero, 2

SLL r4, r0, r1

SLL r5, r2, r3

ADD r6, r4, r5

ADDI r7, \$zero, 666

SEQ r8, r7, r6

SW r3, 0(r5)

LISTA 17, EXERCICIO 5

pos r1, rj

+
rj

sub r1, \$zero, rj

-
rj

lw r1, 0(rj)

MEM
rj

nor r1, rj, \$zero

~
rj

sltiu r1, rj, 1

!
rj

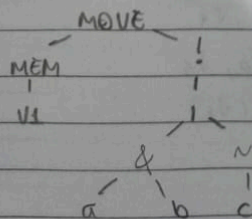
add r1, rj, \$zero

|
rj

addi rj, rj, -1

pre-dec
rj

LISTA 17, EXERCICIO 6 - LETRA A



and r0, a, b

nor r1, c, \$zero

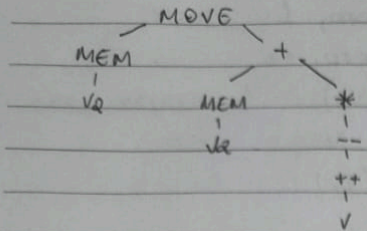
or r2, r0, r1

sltiu r3, r2, 1

sw r3, 0(v1)

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LISTA 17 - EXERCICIO 6 - LETRA B



add r0, v1, \$zero

addi r0, r0, 1

addi r0, r0, -1

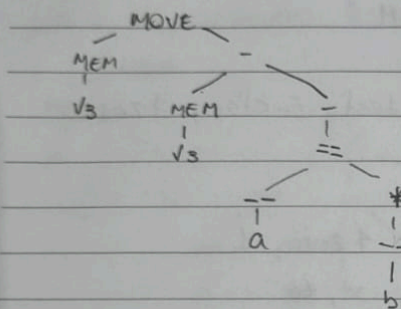
lw r1, 0(r0)

lw r2, 0(v2)

add r3, r2, r1

sw r3, 0(v2)

LISTA 17 - EXERCICIO 6 - LETRA C



add r0, b, \$zero

add r1, a, \$zero

addi r0, r0, -1

addi r1, r1, -1

lw r2, 0(r0)

seq r3, r1, r2

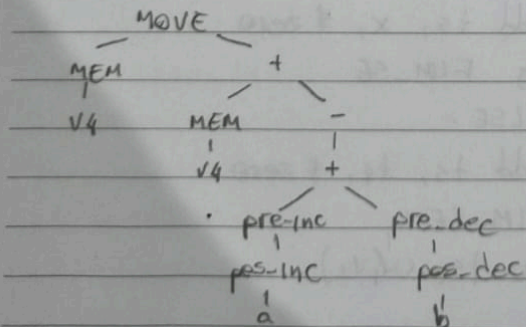
sub r4, \$zero, r3

lw r5, 0(v3)

sub r6, r5, r4

sw r6, 0(v3)

LISTA 17 - EXERCICIO 6 - LETRA D



addi a, a, 1

add r0, a, \$zero

addi a, a, 1

addi b, b, -1

add r1, b, \$zero

addi b, b, -1

add r2, r0, r1

sub r3, \$zero, r2

lw r4, 0(v4)

add r5, r4, r3

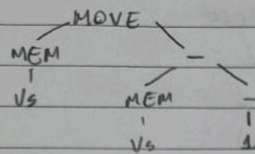
sw r5, 0(v4)

__/__/__

S T Q Q S S D

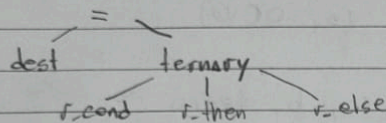
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LISTA 17 - EXERCICIO 6 - LETRA E



```
addi r0, $zero, 1
sub r1, $zero, r0
lw r2, 0(V1)
sub r3, r2, r1
sw r3, 0(V1)
```

LISTA 17 - EXERCICIO 7



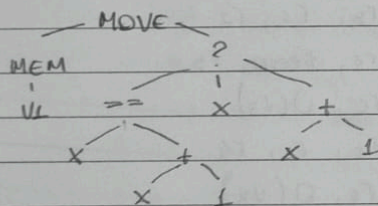
```
beq r-cond, $zero, ELSE
add r-dest, r-then, $zero
j FIM
```

ELSE:

```
add r-dest, r-else, $zero
```

FIM:

LISTA 17 - EXERCICIO 8 - LETRA A



```
addi t0, $zero, 1
```

```
add t1, x, t0
```

```
seq t2, x, t1
```

```
addi t3, $zero, 1
```

```
add t4, x, t3
```

```
beq t2, $zero, ELSE
```

```
add t5, x, $zero
```

```
j FIM-SE
```

ELSE:

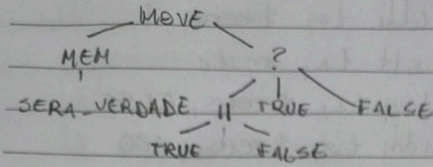
```
add t5, t4, $zero
```

FIM-SE:

```
sw t5, 0(V1)
```


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LISTA 17 - EXERCICIO 8 - LETRA B



or t0, TRUE, FALSE

beq t0, \$zero, ELSE

add t1, TRUE, \$zero

j FIN-SE

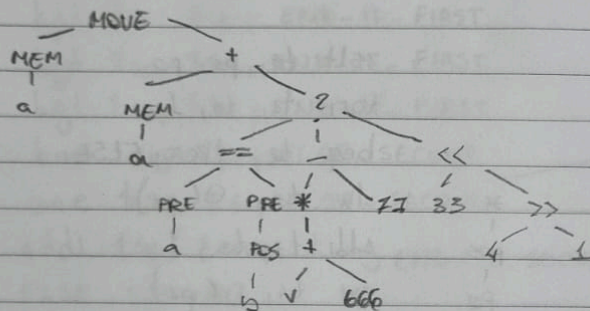
ELSE:

add t1, FALSE, \$zero

FIN-SE:

sw t1, 0(SERA-VERDADE)

LISTA 17 - EXERCICIO 8 - LETRA C



addi v, v, 1

add t0, v, \$zero

addi b, b, -1

add t1, b, \$zero

addi b, b, -1

sub t2, t0, t1

sltu t3, t2, 1

beq t3, \$zero, ELSE

add t4, t2, \$zero

j FIN-SE

ELSE:

add t0, t2, \$zero

FIN-SE:

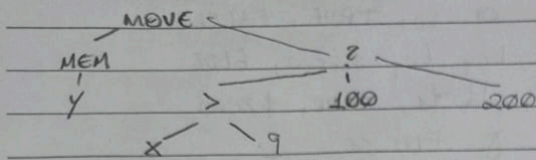
lw t5, 0(a)

add t0, t5, t4

sw t5, 0(a)

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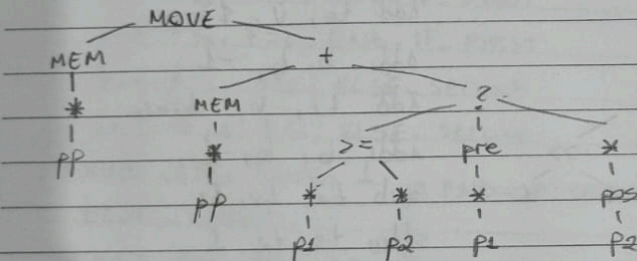
LISTA 17 - EXERCÍCIO 8 - LETRA D



```

addi t0, $zero, 9
slt t1, x, t0
beq t1, $zero, ELSE
addi t2, $zero, 100
j FIM_SE
ELSE:
addi t2, $zero, 200
FIM_SE:
sw t2, 0(y)
  
```

LISTA 17 - EXERCÍCIO 8 - LETRA E



```

slt t0, p1, p2
xorl t1, t0, 1
beq t2, $zero, ELSE
lw t3, 0(p1)
addi t3, t3, 1
sw t4, 0(p1)
add t5, t1, $zero
j FIM_SE
ELSE:
lw t6, 0(p2)
addi p0, p2, -4
FIM_SE:
lw t7, 0(pp)
add t8, t7, t6
sw t8, 0(pp)
  
```


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LISTA 18 - EXERCICIO - 1

addi \$s0, \$zero, -1

j WHILE-TEST

WHILE:

add \$s1, \$v0, \$zero

add \$s2, \$v0, \$zero

add \$s3, \$v0, \$zero

add \$s4, \$s2, \$s3

add \$s5, \$s1, \$s3

add \$s6, \$s1, \$s2

bgt \$s1, \$s4, END-IF-FIRST

bgt \$s2, \$s5, END-IF-FIRST

bgt \$s3, \$s6, END-IF-FIRST

bne \$s1, \$s2, ELSE-SECOND

bne \$s2, \$s3, ELSE-SECOND

addi \$s0, \$zero, 1 j END-IF-SECOND

ELSE-SECOND:

beq \$s1, \$s2, IF-THIRD

beq \$s1, \$s3, IF-THIRD

beq \$s2, \$s3, IF-THIRD

addi \$s0, \$zero, 3

j END-IF-SECOND

IF-THIRD:

addi \$s0, \$zero, 2

END-IF-SECOND:

END-IF-FIRST:

WHILE-TEST:

beq \$s0, -1, WHILE

END-PROGRAM:

return \$s0.

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LISTA 19 - EXERCÍCIO 1 - LETRA A

$$B_1: \text{gen}[B_1] = \{d_1, d_2, d_3, d_4\}$$

$$\text{Kill}[B_1] = \{d_5, d_8\}$$

$$B_2: \text{gen}[B_2] = \{d_5, d_6, d_7\}$$

$$\text{Kill}[B_2] = \{d_1\}$$

$$B_3: \text{gen}[B_3] = \{d_8, d_9, d_{10}\}$$

$$\text{Kill}[B_3] = \{d_2\}$$

$$B_4: \text{gen}[B_4] = \{\}$$

$$\text{Kill}[B_4] = \{\}$$

$$B_5: \text{gen}[B_5] = \{d_{11}, d_{12}, d_{13}, d_{14}, d_{15}, d_{17}, d_{18}\}$$

$$\text{Kill}[B_5] = \{d_{19}, d_{20}\}$$

$$B_6: \text{gen}[B_6] = \{d_{19}, d_{20}, d_{21}, d_{22}, d_{23}, d_{25}, d_{26}\}$$

$$\text{Kill}[B_6] = \{d_{16}, d_{18}\}$$

Bloco	in[B]	out[B]
B1		d_1, d_2, d_3, d_4
B2	$d_1, d_2, d_3, d_4, d_5, d_6$ $d_7, d_8, d_9, d_{10}, d_{11}, d_{12}$	$d_5, d_6, d_7, d_8, d_9, d_{10}, d_{11}, d_{12}$ $d_{18}, d_{19}, d_{20}, d_{21}, d_{22}$
B3	$d_2, \dots, d_{15}, d_{17},$ d_{18}	$d_3, \dots, d_{15}, d_{17},$ d_{18}
B4	$d_3, \dots, d_{15}, d_{17},$ d_{18}	$d_3, \dots, d_{15}, d_{17},$ d_{18}
B5	$d_5, \dots, d_{15}, d_{17}, d_{18}$	$d_3, \dots, d_{15}, d_{17},$ d_{18}
B6	$d_3, \dots, d_{15}, d_{17},$ d_{18}	$d_3, \dots, d_{15}, d_{17}, d_{19},$ $d_{20}, d_{21}, d_{22}, d_{23}, d_{25}, d_{26}$

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LISTA 19 - EJERCICIO 1 - LETRA B

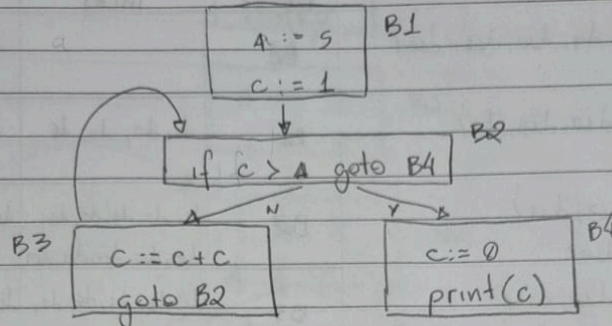
	Blocos	in[B]	out[B]
B0: gen[B0] = {d1, d2, d3, d4}	B0		d1, d2, d3, d4
Kill[B0] = {d6, d7, d8, d9, d10, d11, d16}			
B1: gen[B1] = {d5, d6}	B1	d1, d2, d3, d4	d1, d2, d4, d5, d6
Kill[B1] = {d18, d2, d17, d13, d9}			
B2: gen[B2] = {d7}	B2	d1, d3, d4, d5, d6, d10, d16, d17, d18, d19	d1, d3, d4, d5, d7, d15, d16, d18, d19
Kill[B2] = {d2, d6, d13, d17}			
B3: gen[B3] = {d8, d9, d10}	B3	d1, d3, d4, d5, d6, d10, d16, d17, d18, d19	d1, d3, d7, d8, d9, d10, d15, d16, d18, d19
Kill[B3] = {d3, d4, d12, d14}			
B4: gen[B4] = {d11, d12}	B4	d1, d3, d4, d5, d6, d10, d16, d17, d18, d19	d1, d3, d7, d9, d11, d12, d15, d16, d18, d19
Kill[B4] = {d8, d16, d19}			
B5: gen[B5] = {d13, d14}	B5	d1, d3, d7, d9, d10, d11, d12, d15, d16, d18, d19	d1, d3, d7, d9, d13, d14, d15, d16, d18, d19
Kill[B5] = {d2, d3, d6, d7, d9, d17}			
B6: gen[B6] = {d15, d16}	B6	d1, d3, d4, d5, d6, d10, d16, d17, d18, d19	d1, d3, d7, d9, d13, d15, d16, d18, d19
Kill[B6] = {d1, d11, d19}			
B7: gen[B7] = {d17, d18, d19}	B7	d1, d3, d4, d5, d6, d10, d16, d17, d18, d19	d1, d3, d7, d9, d13, d15, d16, d18, d19
Kill[B7] = {d2, d3, d6, d7, d11, d15, d16}			

LISTA 19 - EJERCICIO 1 - LETRA C

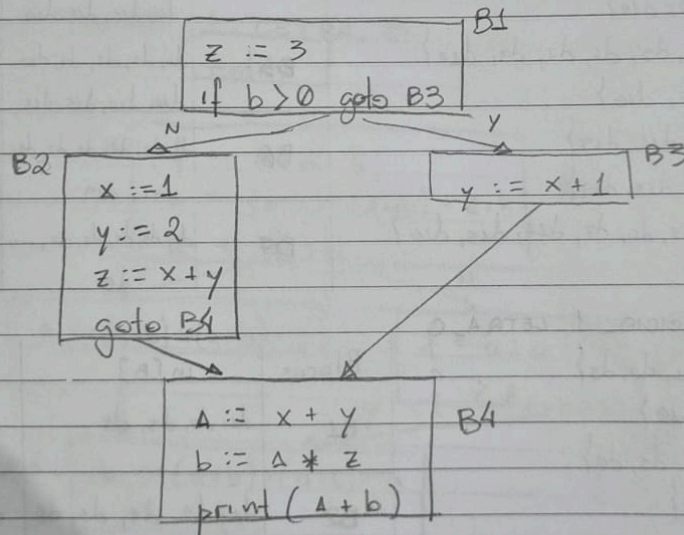
	Blocos	in[B]	out[B]
B1: gen[B1] = {d1, d2, d3}	B1	d1, d2, d3, d4	d1, d2, d4, d6
Kill[B1] = {d10}			
B2: gen[B2] = {d4, d5, d6}	B2	d1, d2, d4, d5, d6	d1, d2, d4, d5, d6, d7, d9
Kill[B2] = {d8}			
B3: gen[B3] = {d7}	B3	d1, d2, d4, d5, d6, d7	d1, d2, d4, d6, d7, d8
Kill[B3] = {}			
B4: gen[B4] = {d8}	B4	d1, d2, d4, d5, d6, d7, d8	d1, d2, d4, d5, d6, d7, d8, d9
Kill[B4] = {d4}			
B5: gen[B5] = {d9}	B5	d1, d2, d4, d5, d6, d7, d8, d9	d1, d2, d4, d6, d7, d8, d9, d10
Kill[B5] = {}			
B6: gen[B6] = {d10}	B6	d1, d2, d4, d5, d6, d7, d8, d9, d10	d1, d3, d4, d5, d8, d9, d10
Kill[B6] = {d2}			

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LISTA 19 - EXERCICIO 2 - LETRA A

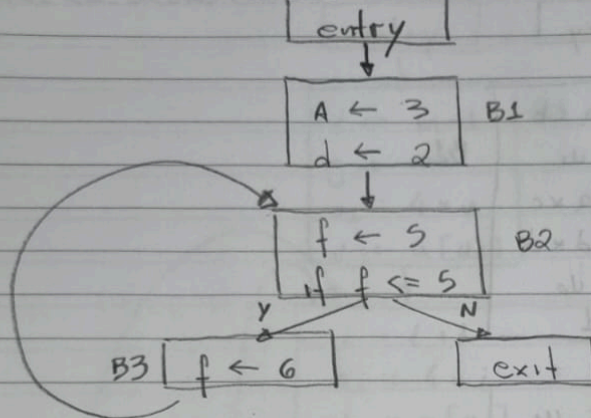


LISTA 19 - EXERCICIO 2 - LETRA B

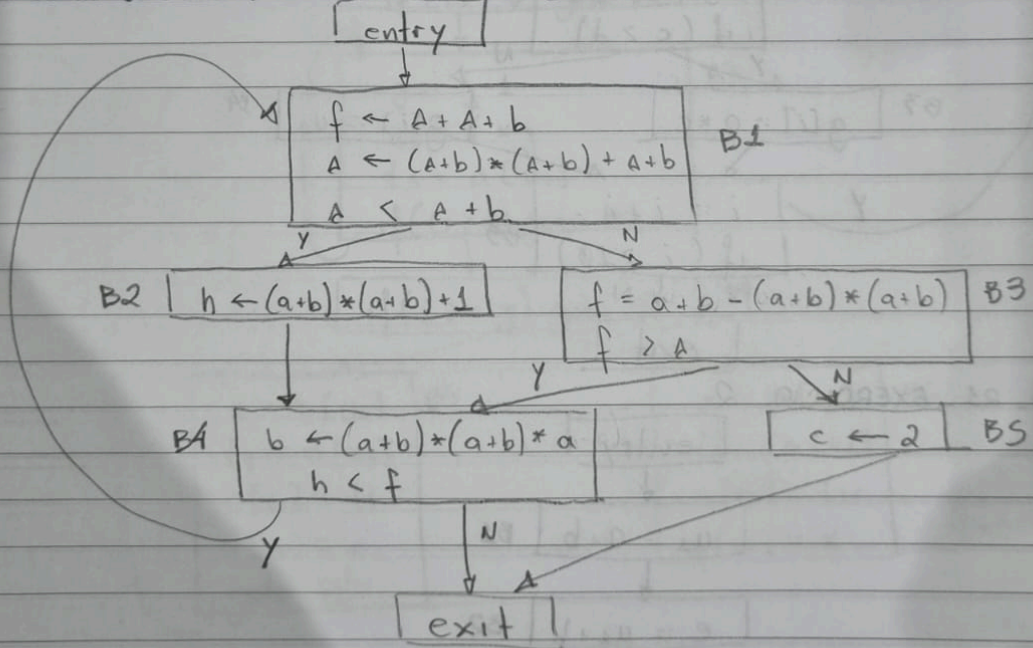


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LISTA 20 - EXERCICIO 1 LETRA A

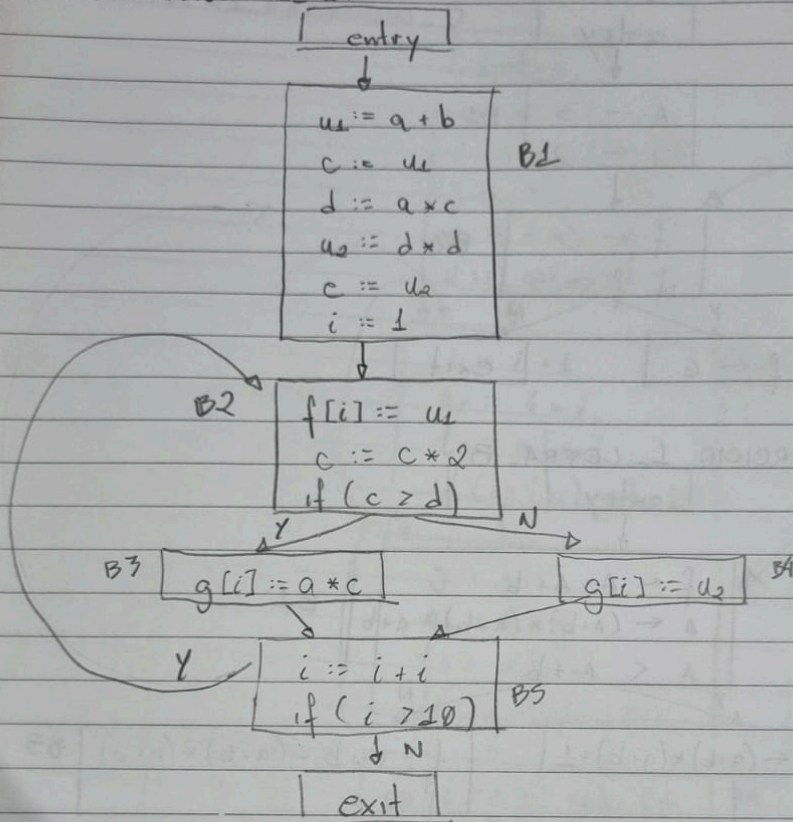


LISTA 20 - EXERCICIO 1 LETRA B

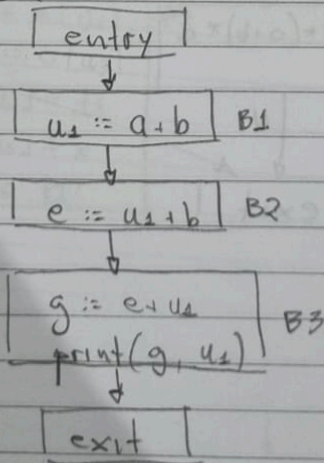


Gabriel Peres de Souza

LISTA 21 - EXERCÍCIO 1



LISTA 21 - EXERCÍCIO 2



LISTA-21-EXERCICIO-3

